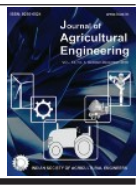
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Convective Drying of Turmeric Rhizome

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ABSTRACT

Experimental trials on convective drying of turmeric (*Curcuma longa* L) rhizomes were conducted using factorial completely randomized design. Turmeric rhizome samples were cured for 30 min in 0.1 % solution of sodium carbonate and dried in a laboratory tray dryer at air temperatures of 50 °C, 60 °C and 70 °C with drying air velocity of 2 $m.s^{-1}$. Drying air temperature was optimized on the basis of maximum retention of curcumin content, and b value of colour with minimum drying time and specific energy consumption. Optimization of drying parameters was carried out using numerical optimization technique. The optimum temperature for convective drying of turmeric rhizomes was found to be 60 °C with drying time of 29 hours.

Turmeric (*Curcuma Longa*), commonly known as *Haldi,* is a primary pigment and is used in diversified forms *viz*. dye, condiment, curry powder, flavouring and colouring agent. Turmeric is also used in preparation of cosmetics, medicinal oils, ointments, *ayurvedic* medicines, and antiseptic creams. It is stomachic, carminative, blood purifier and antiseptic (Jacob, 1995; Anantkawlas, 2014). Cleaning, washing, grading, curing, drying, polishing, packaging and storage are important unit operations involved in turmeric processing. Dry turmeric is utilized for manufacturing of turmeric powder, which is used as flavouring material in food processing (Pruthi, 2006).

The conventional practice of drying turmeric is performed under sun, which is a time consuming method (about 96 h), and produces inferior quality product (Dawn, 1998). Use of appropriate drying technology can produce good quality product. Use of mechanical dryer gives better product quality, avoids the dependency on weather, and reduces microbial contamination of the product (Soysal and Oztekin, 2001).

Presently, drying of agricultural commodities in the food processing industry is being done with various types of mechanical dryer as convective dryer, freeze dryer, infrared dryer, fluidized bed dryer, microwave dryer, and solar dryer. Among these methods of drying, convective drying in convectional tray, cabinet or vacuum dryer is the simplest and most economic method for dehydration of foods, and can be a good solution for drying turmeric rhizomes (Jayaraman and Das Gupta, 1992; Krokida and Marinos-Kouris, 2003). Information on dehydration characteristics of turmeric has been published by researchers (Mulet *et al*., 2005; Lokhande *et al*., 2013; Parveen *et al*., 2013; Borah *et al*., 2015), but little work is reported (Singh *et al.,*2009) on optimization of the drying temperature and drying time for producing quality dried turmeric rhizomes. Therefore, the present study was undertaken with the objective to optimize the process parameters of convective drying to produce dried turmeric rhizomes with the best possible quality in minimum drying time.

MATERIALS AND METHODS

Sample Preparation

Fresh turmeric (var. *Zadol local*) was used for experimental trials. It was procured in bulk from M/S Udaipur Agro Producer Company Ltd., Udaipur. Turmeric rhizomes were washed for 3-5 min in a vegetable washer (capacity 500 $kg.h^{-1}$; Model RPW-500; Bajaj Process Pack Limited, U.P., India) to remove surface adhering dust and dirt. The undesirable portions were removed manually and again washed properly.

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Moisture content of turmeric rhizomes were determined by hot air oven method (Ranganna, 1986). Five samples were taken for determining the initial moisture content of the samples, and average value is reported. Healthy turmeric rhizomes were boiled for 30 min in 0.1% alkaline solution of sodium carbonate to improve colour and characteristic aroma (Sharma *et al.,* 2008; Gagare *et al*., 2015). Boiled rhizomes were taken out and kept in normal water for cooling. Surface water was removed by gentle rubbing with cotton fabric.

Convective Drying

A laboratory model convective tray dryer (Model 12TD, Lab line, Gujrat) having 2 stainless steel trays of 340 × 270 × 38 mm and equipped with 4 electric heaters and thermostat to produce the temperature in the range up to 92 °C was used for drying turmeric rhizomes. The samples (2 kg each) were dried in the tray dryer at 50 °C, 60 °C and 70 °C drying air temperatures. Air drying velocity of 2 $m.s^{-1}$ (Singh *et al.,* 2009) was chosen for dehydration of each sample. Air velocity was measured with a digital anemometer (Swastik Scientific Company, Mumbai; least count 0.1 $m.s^{-1}$), while the correct drying air temperature was adjusted by a temperature indicator-cum-controlling unit. Conditions were allowed to stabilize at a desired temperature. Weight of the sample was recorded at 30 min intervals until a sample attained constant weight. The least count of the weighing scale attached with the dryer was 0.01g. The data were used to analyse the moisture movement characteristics viz. moisture ratio, drying rate, moisture diffusivity, etc.

The moisture ratio (MR) at each moisture content level was determined by the following equation:

$$MR = \frac{M - M_e}{M_o - M_e} \quad \ldots(1)$$

Where,

MR = Moisture ratio,

M = Moisture content at any time, per cent (d.b),

M_o = Initial moisture content, per cent (d.b.), and

Me = Equilibrium moisture content, per cent (d.b.).

Experimental value of the effective diffusivity was calculated by plotting experimental drying data in terms of ln (MR) versus drying time. It gave a straight line, and the slope of the line was used to measure the moisture diffusivity (D_{eff}). The radius of the dried material r_c was assumed to be constant throughout the drying process.

$$Slope = \frac{\beta_{eff}}{r_c^2} \quad \ldots(2)$$

Where,

D_{eff} = Effective diffusivity, $m^2.s^{-1}$,

r_c = Cylinder radius, m, and

β = Roots of the Bessel function = 2.4048.

Quality Evaluation

Food quality is one of the important parameter in food processing to ensure the quality of finished products. Quality of dehydrated turmeric powder was evaluated on the basis of several parameters *viz.* colour and curcumin content. Colour of turmeric powder was measured using a Hunter Lab Colorimeter (Model CLFX-45-2, Spectrum range 400-700 nm; Photometric range 0-150 %; spectral resolution 10 nm). The colour values were obtained as b* value representing yellow (positive *b** value) or blue (negative *b** value) colour (McGuire, 1992).

Curcumin content was determined by solvent extraction, followed by spectrophotometric method (Joshi *et al*., 2009). One hundred milligram of dried rhizome was taken in an extraction flask, 30 ml of 95% alcohol added, and refluxed for 3 h. The refluxed residue was cooled and taken on a filter paper, washed with 95% alcohol, and the volume made up to 100 ml. Two ml of diluted sample was taken in a 25 ml of volumetric flask, the volume made up to 25 ml, and the optical density at 425 nm was measured in a UV-VIS spectrophotometer (Model SL-150, ELICO Limited, Ahmedabad; spectrum range 190-1000 nm; accuracy ±0.5nm, bandwidth 2 nm, readability 0.1nm, repeatability ±0.2nm).

The absorbance value of the standard solution of curcumin (0.00025 g.100 ml^{-1}) was taken as 0.42. The curcumin content in the sample was calculated by using the following formula (Manjunath *et al*., 1991):

$$\text{Curcumin content}(\%) = \frac{0.00025 \times \text{absorbance of sample} \times 100 \times 100}{\text{absorbance of standard} \times \text{weight of sample} \times 5} \quad \ldots(3)$$

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Specific Energy Consumption

Energy consumption in drying was determined in terms of specific energy consumption (SEC), MJ.kg^{-1}, which is defined as the energy consumption per kilogram of water evaporated. The specific energy consumption was estimated for convective drying processes, considering the total energy supplied to dry the turmeric rhizome sample from initial moisture content of about 471.28 % (d.b.) to a desired moisture content of approximately 9.00 % (d.b.). Heat lost through exhaust air was not taken into account, as there was no heat recovery during the drying process. An air blower of 0.5 W and two heaters of 1 kW each were used in the convective dryer and its overall efficiency was assumed to be 90 per cent.

The energy consumption was calculated in terms of specific energy consumption (SEC) by the following equation (Venkatachalapathy, 1998):

$$SEC = \frac{P \times t_{on}(100 - M_f)}{m(M_i - M_f)} \quad \ldots (4)$$

Where,

P = Input power, W,

t_{on} = Total amount of time "on", s,

M_i = Initial moisture content, per cent (w.b.),

M_f = Final moisture content, per cent (w.b.), and

m= Initial mass, kg.

Optimization of Process Parameters

Optimization of the input parameters of drying techniques was done for maximum curcumin content, b value of colour, minimum drying time, and specific energy consumption. The optimization of drying air temperature for convective drying of turmeric rhizome was carried out using numerical optimization technique using Design Expert 8.0.6 statistical software by adopting the factorial completely randomized design. Full second order regression model was fitted in various responses, and independent variables using multiple regression analysis.

RESULTS AND DISCUSSION

The initial moisture content of fresh and cured turmeric rhizomes was found to be 381.93 % (d.b.) and 471.28 % (d.b.), respectively. After drying to nearly constant weight, the moisture content was reduced up to 9.05 % (d.b.) within the permissible limit of 12 % (d.b.) and 10 % (d.b.) prescribed under PFA regulations and Indian standards, respectively (Ravindran *et al*., 2007).

Variation in Moisture Content with Time

The relationship between moisture content and drying time of turmeric rhizome at different temperatures (50 °C, 60 °C and 70 °C) at air velocity of 2 m.s^{-1} are shown in Fig. 1. It can be observed from these curves that the moisture content of turmeric decreased exponentially with drying time. The moisture-time relationship was non-linear, and the drying was faster initially as compared to the later part of drying. Drying time was longest when dried at 50 °C and the least when dried at 70 °C drying temperature. Similar results were reported by Salgado *et al*. (1994) for sugar beet, Prabhanjan *et al.* (1995) for carrot, and Jayashree *et al.* (2012) for ginger. Drying period for turmeric rhizome at 50 °C, 60 °C and 70 °C of drying air temperatures were 36 h, 29 h and 23 h, respectively. It is clear from Table 1 that drying temperature had significant effect on drying time at 1% level of significance.

Table 1. ANOVA for effect of air temperature on drying time

Source	df	SS	MS	F	S_EM	CD (1%)
Temp.	2	294.0	147.0	1764.00**	0.1667	0.577
Error	6	0.5	0.0833		0.1667	
Total	8	294.5				

*** Significant at 1%, CV=0.208*

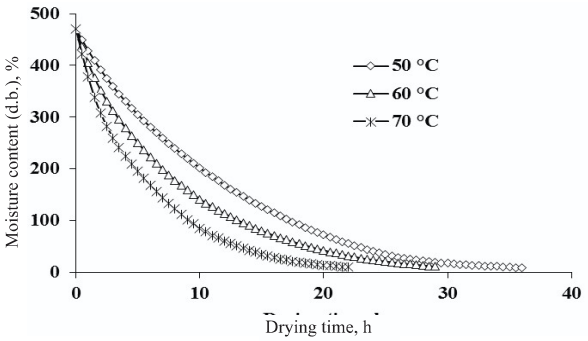


Fig. 1: Variation of moisture content with drying time of turmeric

Drying Rate Curve

The drying rate plotted against the moisture content is presented in Fig. 2. It was found that complete drying

of turmeric took place in the falling rate period, and the constant rate period was completely absent in tray drying. The drying in falling rate period indicated that the internal mass transfer occurred by diffusion. Similar results were reported by Pisal *et al*. (2015) for ginger. It was observed from Fig. 2 that the drying rate was higher in the initial period of drying, and subsequently reduced with decrease in moisture content. Drying rate increased with increase in drying temperature from 50 °C to 70 °C resulting into substantial decrease in drying time. The results are in confirmation with Rocha *et al.* (1992).

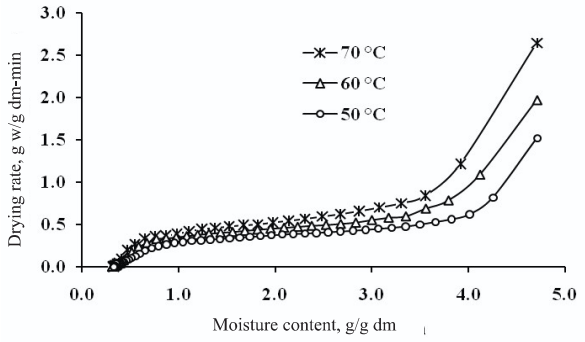


Fig. 2: Variation of drying rate with moisture content

Moisture Diffusivity of Turmeric Rhizome and Slice

The variations in ln (MR) with drying time of turmeric rhizome are presented in Fig. 3. The variation in ln (MR) with drying time for each case was found to be linear with inverse slope. The slope became steeper with increase in drying air temperature. Moisture diffusivities were calculated from the slopes of these straight lines (Doymaz, 2004). Moisture diffusivity was observed to increase with drying air temperature in convective drying processes (Table 2). Results are in confirmation with Rahman and Lamb (1991); Pokharkar and Prasad (1998). Moisture diffusivity

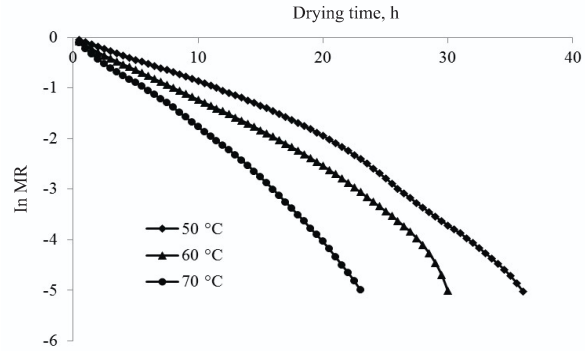


Fig. 3: Variation in ln (MR) with drying time

value ranged from 3.22 $\times 10^{-10}$ to 4.12 $\times 10^{-10}$ $m^2.s^{-1}$, depending upon the drying air temperature. Similar results has also been reported by Lokhande *et al*. (2013); Parveen *et al*. (2013); Borah *et al*. (2015). These diffusivity values are within the general range of 10^{-08} to 10^{-12} $m^2.s^{-1}$ for drying of food materials (McMinn and Magee, 1999).

Table 2. Moisture diffusivity at different drying temperatures

S. No.	Temperature, °C	Diffusivity, $m^2.s^{-1}$	R^2
1	50	3.22039×10^{-10}	0.970
2	60	3.41178×10^{-10}	0.976
3	70	4.11891×10^{-10}	0.987

Quality Evaluation of Convective Dried Turmeric

Quality attributes *viz*. colour b value and curcumin content of dried samples, time required in drying and specific energy consumption during drying for different drying temperature are presented in Table 3.

Table 3. Quality attributes at different drying temperature

S. No.	Drying temp., °C	Colour, b* value	Curcumin content, %	Drying time, h	SEC, $MJ.kg^{-1}$
1.	50	42.91	2.96	36	2161.61
2.	60	44.91	2.92	29	1741.61
3.	70	40.21	2.80	23	1381.04

Colour value

The b value represents the yellowness index of product. The b value of the samples dried at 50 °C, 60 °C and 70 °C was found to be 42.91, 44.91 and 40.21, respectively. The b value of convective dried turmeric samples increased with increase in temperature from 50 °C to 60 °C, but decreased at 70 °C drying temperature. This might be due to adverse effect of high drying temperature on the curcumin.

Analysis of Variance (ANOVA) was applied to study the effect of the variable on the colour (Table 4). Critical difference, 0.0014 and co-efficient of variance (CV) 0.0445 was evaluated for b-value. ANOVA shows that drying temperature had significant effect on colour at 1 % level of significance.

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Table 4. ANOVA for the effect temperature on colour (b-value)

Source	df	SS	MS	F	S_EM	CD (1%)
Temperature	2	25.972	12.9861	24.14**	0.423465	0.0014
Error	6	3.2278	0.53796		0.423465	
Total	8	29.200				

** *Significant at 1%, CV=0.0445*

Curcumin content

Curcumin content of convectivically dried turmeric was found to be 2.96 %, 2.92 % and 2.80 % at drying temperature of 50 °C, 60 °C and 70 °C, respectively. Curcumin content decreased with increasing drying temperature from 50 °C to 70°C. The maximum mean value of curcumin content was at 50 °C, and decreased to 2.80 % when the temperature was increased to 70 °C. Drying temperature had significant effect on curcumin content at 1 % level with CV of 0.0249 (Table 5). The findings of Tiwari *et al*. (2006) are in confirmation that drying air temperature had significant effect on the oleoresin content of dried turmeric.

Specific Energy Consumption

Specific energy consumption during convective drying of turmeric rhizome was found to be 2161.61 MJ.kg^{-1}, 1741.61 MJ.kg^{-1}, and 1381.04 MJ.kg^{-1} at drying temperatures of 50 ºC, 60 ºC and 70 °C, respectively. Maximum specific energy consumption was recorded at 50 ºC because of longer drying time requirement, while minimum specific energy consumption was at 70ºC of drying temperature because of low drying time. The results are in agreement with the findings of Aghbashlo *et al.* (2008).

Optimization of Process Parameters

Maximum curcumin content and colour (b value), minimum drying time and specific energy consumption were considered for optimization of drying parameters. The desired goals for each factor and response are shown in Table 6. In order to optimize the process parameters for convective drying by numerical optimization, which finds a point that maximizes the desirability function; equal importance of '3' was given to the process parameters and 5 responses. The numerical optimization gave optimum value of drying air temperature as 60.82 °C, 28.15 h drying time for 42.43 % colour b value, and 2.88 % curcumin content, respectively, for turmeric rhizome.

CONCLUSIONS

Moisture diffusivity of turmeric rhizome increased with increase in drying air temperature, and varied in the range 3.2204 ×10^{-10} to 4.1189 ×10^{-10} m^2.s^{-1} depending upon the drying air temperatures. Drying temperature

Table 5. ANOVA for effect of temperature on curcumin content of turmeric powder

Source	df	SS	MS	F	S_EM	CD (1%)
Temperature	2	0.040909	0.020455	146.78**	0.006815	0.0236
Error	6	0.000836	0.000139		0.006815	
Total	8	0.041746				

** *Significant at 1%, CV=0.0249*

Table 6. Process variable and responses for convective drying

Attribute	Goal	Lower limit	Upper limit	Importance
Temperature, °C	in range	50	70	3
b* value	Maximize	40.21	44.91	3
Curcumin content, %	Maximize	2.80	2.96	3
Drying time, h	Minimize	23	36	3
SEC, MJ.kg^{-1}	Minimize	1381.03	2161.61	3

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had significant effect on curcumin content and colour of dried products. Curcumin content of convective dried turmeric rhizome decreased from 2.96 % to 2.80 %, with increase in drying temperature from 50 ºC to 70 °C. The b values of convective dried turmeric rhizome ranged from 42.91 to 40.21. Specific energy consumption was found to be in the range 2161.61 MJ.kg^{-1} to 1381.04 MJ.kg^{-1}, and decreased with reduced drying time with increase in drying temperature. The optimum drying temperature was found to be 60 °C for convective drying of turmeric rhizomes with drying time of 23 hours.

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